

ONE WHOLE		= 1 = 100%	$\frac{1}{1}$
ONE HALF		= 0.5 = 50%	$\frac{1}{2}$ $\frac{1}{2}$
ONE THIRD		= 0.333 = 33.3%	$\frac{1}{3}$ $\frac{1}{3}$ $\frac{1}{3}$
ONE FOURTH		= 0.25 = 25%	$\frac{1}{4}$ $\frac{1}{4}$ $\frac{1}{4}$ $\frac{1}{4}$
ONE FIFTH		= 0.20 = 20%	$\frac{1}{5}$ $\frac{1}{5}$ $\frac{1}{5}$ $\frac{1}{5}$ $\frac{1}{5}$
ONE SIXTH		= 0.166 = 16.6%	$\frac{1}{6}$ $\frac{1}{6}$ $\frac{1}{6}$ $\frac{1}{6}$ $\frac{1}{6}$ $\frac{1}{6}$
ONE SEVENTH		= 0.142 = 14.2%	$\frac{1}{7}$ $\frac{1}{7}$ $\frac{1}{7}$ $\frac{1}{7}$ $\frac{1}{7}$ $\frac{1}{7}$ $\frac{1}{7}$
ONE EIGHTH		= 0.125 = 12.5%	$\frac{1}{8}$ $\frac{1}{8}$ $\frac{1}{8}$ $\frac{1}{8}$ $\frac{1}{8}$ $\frac{1}{8}$ $\frac{1}{8}$ $\frac{1}{8}$
ONE NINTH		= 0.111 = 11.1%	$\frac{1}{9}$ $\frac{1}{9}$ $\frac{1}{9}$ $\frac{1}{9}$ $\frac{1}{9}$ $\frac{1}{9}$ $\frac{1}{9}$ $\frac{1}{9}$ $\frac{1}{9}$
ONE TENTH		= 0.10 = 10%	$\frac{1}{10}$ $\frac{1}{10}$ $\frac{1}{10}$ $\frac{1}{10}$ $\frac{1}{10}$ $\frac{1}{10}$ $\frac{1}{10}$ $\frac{1}{10}$ $\frac{1}{10}$ $\frac{1}{10}$



★ $14.28\% = \frac{1}{7}$

Initial : Final

$\frac{7}{8}$
+ 1 (14.28% of 7)

$\boxed{7 : 8}$

* $700 \xrightarrow[+ 14.28\% \text{ of } 700]{+ 100} 800$

$\left\{ \begin{array}{l} 700 : 800 \\ 7 : 8 \end{array} \right\} \quad 700 \times \frac{1}{7} = 100$

* $1400 \xrightarrow[+ 14.28\% \text{ of } 1400]{+ 200} 1600$

$\left\{ \begin{array}{l} 1400 : 1600 \\ 14 : 16 \\ 7 : 8 \end{array} \right\} \quad 1400 \times \frac{1}{7} = 200$

* $770 \xrightarrow[+ 14.28\% \text{ of } 770]{+ 110} 880$

$\left\{ \begin{array}{l} 770 : 880 \\ 77 : 88 \\ 7 : 8 \end{array} \right\} \quad 770 \times \frac{1}{7} = 110$

Decrease :-

$$14.28\% = \frac{1}{7}$$

Initial : Final

7

6

-1 (14.28% of 7)

7:6

* 840 $\xrightarrow[-14.28\% \text{ of } 840]{-120}$ 720 $\left(840 \times \frac{1}{7} = 120\right)$

$$\left\{ \begin{array}{l} 840:720 \\ 84:72 \\ 7:6 \end{array} \right\}$$

* 560 $\xrightarrow{-80}$ 480
 -14.28%
 of 560 $\left(560 \times \frac{1}{7} = 80\right)$

$$\left\{ \begin{array}{l} 560 : 480 \\ 56 : 48 \\ 7 : 6 \end{array} \right\}$$

* 490 $\xrightarrow[-14.28\%]{-70}$ 420 $\left(490 \times \frac{1}{7} = 70\right)$

$$\left\{ \begin{array}{l} 490 : 420 \\ 49 : 42 \\ 7 : 6 \end{array} \right\}$$

Find the ratio of Initial and Final Number $\rightarrow$
 $+$ $\Rightarrow$ Increase
 $-$ $\Rightarrow$ Decrease

$+$ $\Rightarrow$ Increase

$\Rightarrow$ Decrease

• + 25% $\rightarrow$ $\frac{I:F}{4:5}$
(1/4) $\downarrow$
+1

• $+ 28 \cdot 56 \cdot \dots \rightarrow$

$(2/7)$

$I : F$

$7 : 9$

$\rightarrow +2$

$$\bullet -25 \cdot \frac{1}{4} \rightarrow \begin{matrix} 4 & 3 \\ & \underbrace{-1} \end{matrix}$$

$$\bullet -28.56 \cdot \rightarrow \begin{matrix} 7:5 \\ -2 \end{matrix}$$

$$\bullet + 12.5 \div \rightarrow \begin{array}{r} 8:9 \\ \hline +1 \end{array}$$

• $+ 16.66\% \rightarrow$ $6:7$
 $(1/6)$ $\xrightarrow{+1}$

$$\bullet - 12 \cdot 5 \cdot \frac{1}{8} \rightarrow \begin{array}{r} 8 \cdot 7 \\ -1 \end{array}$$

• $-16.66\% \rightarrow$ $\begin{matrix} 6 & : & 5 \\ & \text{---} & \\ & -1 & \end{matrix}$

$$\bullet + 25\% \Rightarrow \begin{array}{l} I:F \\ 4:5 \end{array} \Rightarrow \text{Also, } \begin{array}{l} I:F \\ 1:\frac{5}{4} \end{array}$$

$$\Rightarrow \frac{I}{F} = \frac{4}{5} \quad \boxed{\frac{5}{4} I = F}$$

These are
the ways
of
Representation

$$\bullet - 25\% \Rightarrow \begin{array}{l} I:F \\ 4:3 \end{array} \Rightarrow \text{Also, } \begin{array}{l} I:F \\ 1:\frac{3}{4} \end{array}$$

$$\Rightarrow \frac{I}{F} = \frac{4}{3} \quad \boxed{\frac{3}{4} I = F}$$

$$\bullet + 20\% \Rightarrow \frac{1}{5} \quad \begin{array}{l} I:F \\ 5:6 \end{array} \Rightarrow F = I \times \frac{6}{5} \quad \left(1 + \frac{1}{5} = \frac{6}{5}\right)$$

$$\bullet + 10\% \Rightarrow \frac{1}{10} \quad \begin{array}{l} I:F \\ 10:11 \end{array} \Rightarrow F = I \times \frac{11}{10} \quad \left(1 + \frac{1}{10} = \frac{11}{10}\right)$$

$$\bullet + 14.28\% \Rightarrow \frac{1}{7} \quad \begin{array}{l} I:F \\ 7:8 \end{array} \Rightarrow F = I \times \frac{8}{7} \quad \left(1 + \frac{1}{7} = \frac{8}{7}\right)$$

$$\bullet + 16.66\% \Rightarrow \frac{1}{6} \quad \begin{array}{l} I:F \\ 6:7 \end{array} \Rightarrow F = I \times \frac{7}{6} \quad \left(1 + \frac{1}{6} = \frac{7}{6}\right)$$

$$\bullet + 25\% \Rightarrow 1 + \frac{1}{4} = \frac{5}{4} \Rightarrow I + \frac{I}{4} = F$$

$$I \left(1 + \frac{1}{4}\right) = F$$

$$* \quad \boxed{I \times \frac{5}{4} = F}$$

When Initial Number
is increased by 25%

$$\cdot + 6.25\% \Rightarrow \frac{1}{16}$$

$$I : F \\ 16 : 17$$

$$I \times \frac{17}{16} = F$$

$$\cdot + 6.66\% \Rightarrow \frac{1}{15}$$

$$I : F \\ 15 : 16$$

$$I \times \frac{16}{15} = F$$

$$\cdot + 11.11\% \Rightarrow \frac{1}{9}$$

$$I : F \\ 9 : 10$$

$$I \times \frac{10}{9} = F$$

$$\cdot + 9.09\% \Rightarrow \frac{1}{11}$$

$$I : F \\ 11 : 12$$

$$I \times \frac{12}{11} = F$$

$$\cdot + 25\% \Rightarrow \frac{1}{4}$$

$$I : F \\ 4 : 5$$

$$I \times \frac{5}{4} = F$$

$$\cdot - 6.25\% \Rightarrow \frac{1}{16}$$

$$I : F \\ 16 : 15$$

$$I \times \frac{15}{16} = F$$

$$\cdot - 6.66\% \Rightarrow \frac{1}{15}$$

$$I : F \\ 15 : 14$$

$$I \times \frac{14}{15} = F$$

$$\cdot - 11.11\% \Rightarrow \frac{1}{9}$$

$$I : F \\ 9 : 8$$

$$I \times \frac{8}{9} = F$$

$$\cdot - 9.09\% \Rightarrow \frac{1}{11}$$

$$I : F \\ 11 : 10$$

$$I \times \frac{10}{11} = F$$

$$\cdot - 25\% \Rightarrow \frac{1}{4}$$

$$I : F \\ 4 : 3$$

$$I \times \frac{3}{4} = F$$

$$\cdot + 10\% \Rightarrow \frac{1}{10}$$

$$I : F \\ 10 : 11$$

$$I \times \frac{11}{10} = F$$

$$\cdot - 10\% \Rightarrow \frac{1}{10}$$

$$I : F \\ 10 : 9$$

$$I \times \frac{9}{10} = F$$

• Increase 480 by 25% $\Rightarrow 25\% = \frac{1}{4}$

1) $480 + 120 = 600$

2) $480 \times \frac{5}{4} = 600$

$$\left\{ 1 + \frac{1}{4} = \frac{4+1}{4} = \frac{5}{4} \right\}$$

• Increase 600 by 20% $\Rightarrow 20\% = \frac{1}{5}$

$$600 \times \frac{6}{5} = 720$$

• Increase 500 by 10% $\Rightarrow 10\% = \frac{1}{10}$

$$500 \times \frac{11}{10} = 550$$

• Increase 880 by 12.5% $\Rightarrow 12.5\% = \frac{1}{8}$

$$880 \times \frac{9}{8} = 990$$

• Increase 666 by 16.66% $\Rightarrow 16.66\% = \frac{1}{6}$

$$666 \times \frac{7}{6} = 777$$

* Ways of Representation $\rightarrow$

$$\Rightarrow \frac{a}{b} \div \frac{c}{d}$$

$$ad : cb$$

$$\Rightarrow \frac{3}{5} : \frac{4}{7}$$

$$(3 \times 7) : (5 \times 4)$$

$$21 : 20$$

$$\Rightarrow 5 : 7$$

$$(5 \times 1) : 7$$

$$5 : 7$$

$$\Rightarrow \frac{9}{11} : \frac{3}{5}$$

$$(9 \times 5) : (11 \times 3)$$

$$45 : 33$$

$$15 : 11$$

A bag weighs 80 kg. It is increased by 25%. Find the new weight.

एक बैग का वजन 80 kg है। इसमें 25% की वृद्धि की जाती है। नया वजन ज्ञात कीजिए।

$$1) \quad 80 \xrightarrow{+20} \underline{\underline{100}}$$

$+25\% \text{ of } 80$

$$2) \quad F = I \times \frac{5}{4} \quad F = 80 \times \frac{5}{4} \left(1 + \frac{1}{4}\right)$$

$$F = \underline{\underline{100}}$$

$$3) \quad \begin{array}{l} I : F \\ 4 : 5 \\ \times 20 \quad \left(\begin{array}{l} 80 : 100 \end{array} \right) \times 20 \end{array}$$

These are the methods or alternatives to solve the question.

The price of a pen is 40. It is decreased by 10%. Find the new price.

एक पेन की कीमत 40 है। इसकी कीमत में 10% की कमी की जाती है। नई कीमत ज्ञात कीजिए।

$$1) \quad 40 \xrightarrow{-4} \underline{\underline{36}}$$

$-10\% \text{ of } 40$

$$3) \quad \begin{array}{l} I : F \\ \cancel{10} : 9 \quad (\text{Decrease by } 10\%) \\ \downarrow \quad \downarrow \times 4 \\ \cancel{40} \quad \underline{\underline{36}} \end{array}$$

$$2) \quad F = I \times \frac{9}{10}$$

Decrease by 10%.

$$F = 40 \times \frac{9}{10} = \underline{\underline{36}}$$

A student scored 160 marks. Marks are decreased by 12.5%. Find the new marks.

एक छात्र ने 160 अंक प्राप्त किए। अंकों में 12.5% की कमी की जाती है। नए अंक ज्ञात कीजिए।

$$1) \quad 160 \xrightarrow{-20} \underline{\underline{140}}$$

$-12.5\% \text{ of } 160 \quad \left(160 \times \frac{1}{8} = 20\right)$

$$3) \quad \begin{array}{l} I : F \\ \cancel{8} : 7 \\ \downarrow \quad \downarrow \times 20 \\ \cancel{160} \quad \underline{\underline{140}} \\ 20 \end{array}$$

$$2) \quad F = 160 \times \frac{7}{8}$$

$$F = \underline{\underline{140}}$$

The weight of a packet increases from 48 kg to 60 kg. Find the percent increase.

एक पैकेट का वजन 48 kg से बढ़कर 60 kg हो जाता है। प्रतिशत वृद्धि ज्ञात कीजिए।

1) $48 \xrightarrow{+12} 60$

(Increased by 12)

$$\frac{12}{48} \times 100 = \frac{100}{4}$$

$$= \underline{\underline{25\%}}$$

3) I : F

~~48~~ : ~~60~~

4 : 5

$\xrightarrow{-1}$

$$\frac{1}{4} \times 100 = 25\%$$

2) $48 \longrightarrow 60$

$$\frac{60}{48} = \frac{5}{4}$$

$$\frac{5}{4} - 1 = \frac{1}{4} \text{ or } \underline{\underline{25\%}}$$

The marks of a student decrease from 96 to 84. Find the percent decrease.

एक छात्र के अंक 96 से घटकर 84 हो जाते हैं। प्रतिशत कमी ज्ञात कीजिए।

1) $96 \xrightarrow{-12} 84$

(Decreased by 12)

$$\frac{12}{96} \times 100 = \frac{100}{8}$$

$$= \underline{\underline{12.5\%}}$$

3) I : F

~~96~~ : ~~84~~

8 : 7

$\xrightarrow{-1}$

$$\frac{1}{8} \times 100 = \underline{\underline{12.5\%}}$$

2) $96 \longrightarrow 84$

$$\frac{84}{96} = \frac{7}{8}$$

$$1 - \frac{7}{8} = \frac{1}{8}$$

The number of employees decreases from 360 to 300. Find the percent decrease.

कर्मचारियों की संख्या 360 से घटकर 300 हो जाती है। प्रतिशत कमी ज्ञात कीजिए।

$$1) \quad 360 \xrightarrow{-60} 300$$

(Decreased by 60)

$$\frac{60}{360} \times 100 = 16.66\%$$

$$2) \quad 360 \longrightarrow 300$$

$$\frac{300}{360} = \frac{5}{6}$$

$$1 - \frac{5}{6} = \frac{1}{6} \text{ or } 16.66\%$$

$$3) \quad I : F$$

$$360 : 300$$

$$36 : 30$$

$$6 : 5$$

$$\begin{array}{c} \text{---} \\ -1 \end{array}$$

$$\frac{1}{6} \times 100 = 16.66\%$$

After a 16.67% increase, the marks become 105. Find the original marks.

16.67% की वृद्धि के बाद अंक 105 हो जाते हैं। मूल अंक ज्ञात कीजिए।

$$1) \quad 16.67\% = \frac{1}{6}$$

$$+ 16.67\% = 1 + \frac{1}{6} = \frac{7}{6}$$

$$F = I \times \frac{7}{6}$$

$$105 = I \times \frac{7}{6} \Rightarrow \boxed{I = 90}$$

$$2) \quad I : F$$

$$6 : 7$$

$$\times 15$$

$$\textcircled{90}$$

$$\textcircled{105}$$

$$15$$

A earns ₹ 56000 and B earns ₹ 40000. A's salary is how much percent more than B's salary?

A की आय ₹ 56000 है और B की आय ₹ 40000 है। A की आय, B की आय से कितने प्रतिशत अधिक है?

$$1) \quad A : B$$

$$56000 : 40000$$

$$+ 16000$$

$$\frac{16000}{40000} \times 100 = 40\%$$

$$2) \quad A : B$$

$$56000 : 40000$$

$$56 : 40$$

$$7 : 5$$

$$\textcircled{+2}$$

$$\frac{2}{5} \times 100 = 40\%$$

Q. B's salary is how much percent less than A's?

$$A : B$$

$$56000 : 40000$$

$$7 : 5$$

$$\text{---} -2$$

$$\Rightarrow \frac{2}{7} \times 100 = 28.56\%$$

Q. B's salary is what percent of A's salary?

$$\frac{5}{7} \times 100 \Rightarrow 71.42\%$$

Q. A's salary is what percent of B's salary?

$$\frac{7}{5} \times 100 \Rightarrow 140\%$$

Rahul scored 72 and Amit scored 90. Rahul's marks are how much percent less than Amit's marks?

राहुल ने 72 अंक और अमित ने 90 अंक प्राप्त किए। राहुल के अंक, अमित के अंकों से कितने प्रतिशत कम हैं?

Rahul : Amit

$$72 : 90$$

$$4 : 5$$

$$\text{---} -1$$

$$\frac{1}{5} \times 100 \Rightarrow 20\%$$

Base value will be
Amit's marks.

Two quantities are 64 and 24. Their difference is what percent of the smaller quantity?

दो राशियाँ 64 और 24 हैं। उनका अंतर, छोटी राशि का कितने प्रतिशत है?

Quantity 1 : Quantity 2

$$64 : 24$$

$$8 : 3$$

$$5 \leftarrow \text{Difference}$$

Difference

$$\frac{5}{3} \times 100$$

smaller qty.

$$\left(\frac{3}{3} + \frac{2}{3} \right) \Rightarrow 1 + \frac{2}{3}$$

$$\downarrow \qquad \qquad \downarrow$$

$$100\% \qquad 66.66\%$$

$$\frac{5}{3} \Rightarrow 166.66\%$$

A is 24 and B is 64. By what percent should A increase to become equal to B?

A का मान 24 है और B का मान 64 है। A को B के बराबर होने के लिए कितने प्रतिशत बढ़ाना होगा?

$$A : B$$

$$\cancel{24} : \cancel{64}$$

$$3 : 8$$

$$\begin{array}{c} \text{---} \curvearrowright \text{---} \\ +5 \end{array}$$

{ 3 should be increased by 5 }
to become equal to 8

$$\frac{5}{3} \times 100 \Rightarrow \left(\frac{3+2}{3} \right) \Rightarrow 1 + \frac{2}{3}$$

$$\Rightarrow \underline{\underline{166.67\%}}$$

Two students scored 84 and 105. By what percent should 84 increase to become equal to 105?

दो छात्रों ने 84 और 105 अंक प्राप्त किए। 84 को 105 के बराबर होने के लिए कितने प्रतिशत बढ़ाना होगा?

Student 1 : Student 2

$$\cancel{84} : \cancel{105}$$

$$28 : 35$$

$$\begin{array}{c} \text{---} \curvearrowright \text{---} \\ +7 \end{array}$$

{ 28 should be increased }
by 7 to become 35

$$1 \frac{7}{28} \times 100 = \frac{1}{4} \times 100$$

$$= 25\%$$

A number is increased by 25% to become 18,375. Find the original number.

एक संख्या को 25% बढ़ाने पर उसका मान 18,375 हो जाता है। मूल संख्या ज्ञात कीजिए।

(A) 13,500

(B) 14,500

(C) 15,000

(D) 14,700

(E) 12,700

$$1) \quad 25\% = \frac{1}{4}$$

$$+ 25\% = 1 + \frac{1}{4} = \frac{5}{4}$$

$$F = I \times \frac{5}{4}$$

$$18375 = I \times \frac{5}{4} \Rightarrow \boxed{I = 14700}$$

$$2) \quad I : F$$

$$4 : \cancel{5}$$

$$\begin{array}{c} \times 3675 \\ \downarrow \quad \downarrow \end{array}$$

$$\begin{array}{r} 14700 \quad \cancel{18375} \\ \hline 3675 \end{array}$$

A number is increased by 15.38% and the resultant number is 3,225. What is the original number?

एक संख्या को 15.38% बढ़ाने पर प्राप्त संख्या 3,225 है। मूल संख्या ज्ञात कीजिए।

(A) 2,975

(B) 2,925

(C) 2,795

(D) 2,800

(E) 2,745

$$1) \quad 15.38\% = \frac{2}{13}$$

$$+ 15.38\% = 1 + \frac{2}{13} = \frac{15}{13}$$

$$3225 = I \times \frac{15}{13}$$

$$I = \frac{3225 \times 13}{15}$$

$$I = 2795$$

$$2) \quad I : F \\ 13 : 15$$

$\times 215$

$$\underline{2795} \quad 3225$$

If 4,158 is decreased by 16.67%, what is the resulting number?

यदि 4,158 में 16.67% की कमी की जाए, तो प्राप्त संख्या क्या होगी?

(A) 3,450

(B) 3,465

(C) 3,465.5

(D) 3,665

(E) 3,900

$$1) \quad 4158 \quad \begin{array}{r} - 693 \\ - 16.67\% \\ \hline \end{array} \rightarrow 3465$$

of 4158

$$\left(4158 \times \frac{1}{6} = 693 \right)$$

$$2) \quad F = I \times \frac{5}{6}$$

$$F = 4158 \times \frac{5}{6}$$

$$F = \underline{3465}$$

$$3) \quad I : F \\ \cancel{6} : 5$$

$\times 693$

$$\cancel{4158} \\ 693$$

$$\underline{3465}$$